



sim2claw

training data robots can trust

evidence-backed simulation-to-robot learning for generalist robots

AITX Community x NVIDIA Claw Agent Hackathon · July 17-19

LIVE DEMO · BOOTH 30 · github.com/jakekinchen/sim2claw

Generalist robots are ready to learn. The data isn't.

- VLA / foundation robot models (GROOT, pi-0, RT-X) are starving for manipulation data
- Real teleoperation does not scale: thousands of hours, one scene at a time
- Scraped demos blur 'we demonstrated once' with 'the policy generalized'
- Result: impressive demos, unverifiable training data

the bottleneck is trustworthy data - not model capacity

sim2claw: an evidence-grade data engine

CAPTURE

any real workcell -> 3DGS scan + measured layout, in minutes

TWIN

reviewed MuJoCo physics workcell: board, pieces, two S0-101 arms

TEACH

humans demonstrate a few grasp styles at 20 Hz - not every task

MULTIPLY

simulation retargets them combinatorially into whole datasets

PROVE

a separate frozen evaluator grades every skill - receipts included

"teleoperate grasp styles, not task instances - generate instances combinatorially in simulation"

what you're watching in the live demo

Calibration

orbit the 334,537-splat Gaussian capture of the real workcell

Record

20 Hz leader/follower teleoperation -> source episodes + receipts

Replay

MuJoCo state replay vs synchronized camera recordings, side by side

Evaluator

independent pass/fail verdicts - watch a graded Success live

Library

24 replayable evidence episodes, hash-bound, inspection-only

Studio is read-only evidence - nothing on screen can overwrite a verdict

the data product

LeRobot v2.1

Parquet + MP4 · 8,712 frames · multi-view RGB + actions + language

- overhead C922 + side Logitech + wrist D405 - synchronized multi-camera streams
- 24 deterministic language prompts mapped to 12 move semantics
- every episode hash-bound with a versioned JSON receipt
- failures preserved as counterexamples - never relabeled as success
- sealed held-outs: zero test leakage into training

the format robot foundation models train on - with provenance no scrape has

our data already trains a real VLA



- training completed end-to-end on Brev A100s from our export - pipeline proven
- development rollout was a graded negative (0 mm lift) - reported with full receipts
- that honesty is the point: negatives become reusable counterexample data

evidence, not vibes

94.88 mm

held-out rook lift by a 957K-param ACT policy
trained on a laptop - graded by a separate CPU evaluator

94.4%

simulator error reduction across 8 revisions
309.6 mm -> 17.4 mm event-fit RMS (chart at right)

SIMULATOR ACCURACY

94.4 percent spatial-error reduction across seven stages

Sequential simulator spatial-error progression

Training-side event RMS - millimeters - lower is better

TOTAL ERROR REDUCTION
94.4%

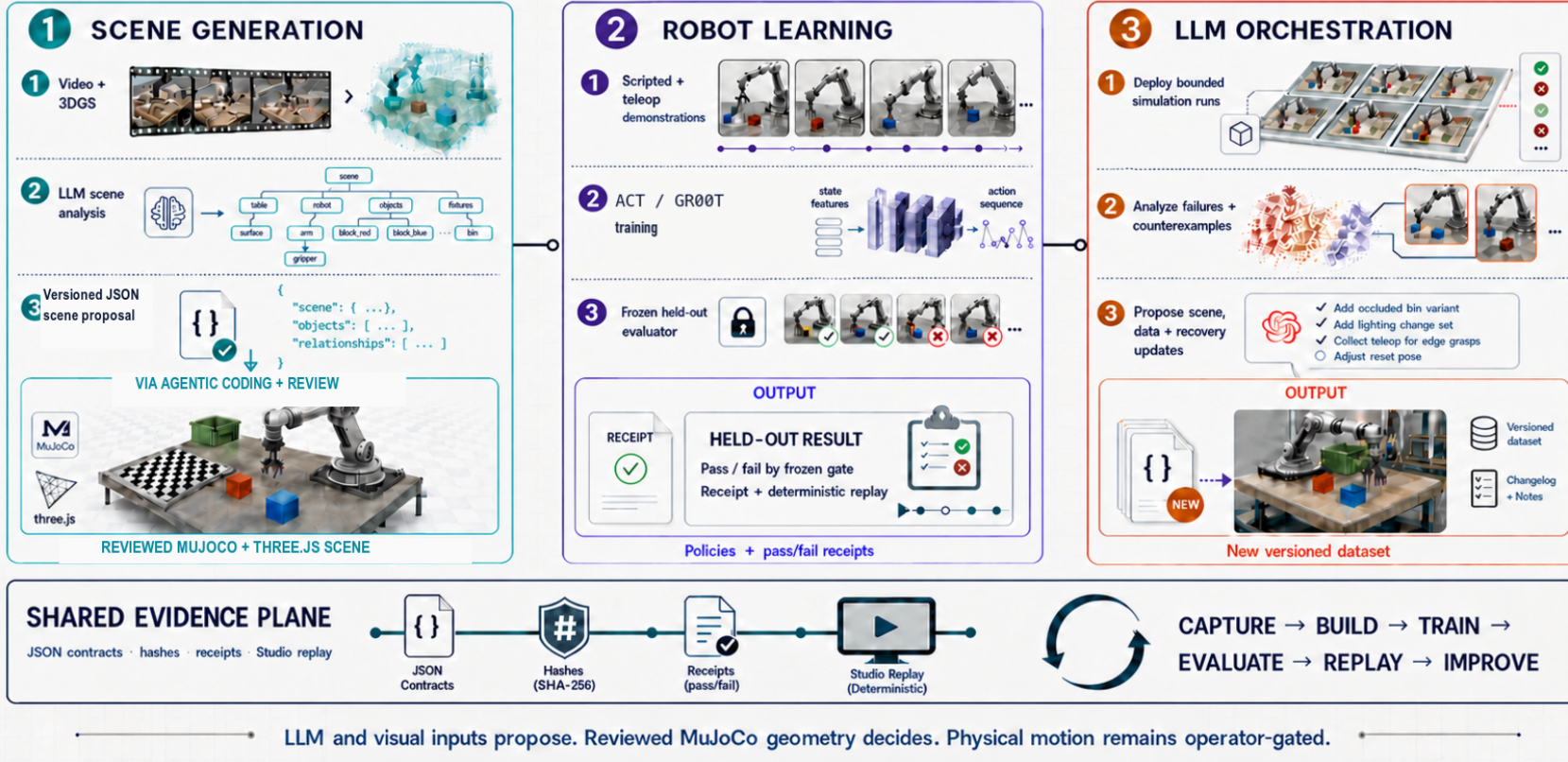


five evidence classes - simulation / replay / learned-policy / physical-read-only / physical-task - never conflated

under the hood

sim2claw — Three systems, one evidence loop

The simple flow expands into three reviewable engines.



MuJoCo 3.10 · SO-101 arms · ACT (PyTorch/MPS) · GR00T N1.7 (Brev A100) · 3DGS (COLMAP/Brush) · Studio (Three.js/Spark)

recursion all the way down

- the repo itself was built by a supervised AI agent loop: Executor -> Reviewer -> Manager
- every slice audited: deterministic tests, session logs, scoped commits
- the learning factory closes the skill loop: calibrate -> train -> mine failures -> re-exam
- failures feed back as targeted correction demos - the exam never gets easier
- the trainer never grades itself - a separately-owned evaluator owns pass/fail

a system that gets measurably smarter every cycle - and can prove it



come see it live - BOOTH 30

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